Renormalisation of Wilson Loops in Four-Dimensional Yang–Mills Theory

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Bałaban's renormalisation group (RG) construction gives a gauge-fixed lattice measure and a corresponding RG trajectory for Yang–Mills at every finite ultraviolet cutoff. However, the gauge transformations introduced at successive steps accumulate, and seem to hinder convergence of the resulting gauge-fixed fields as the cutoff is removed.[1]

We therefore seek the continuum theory through gauge-invariant observables. The natural candidates are Wilson loops and pointwise traces of powers of the field strength, which present a second difficulty: in dimensions greater than two, they are ultraviolet singular and cannot be treated as ordinary continuum random variables.

Our programme addresses this problem for correlation functions of Wilson loops in four-dimensional Yang–Mills theory. We aim to identify the scalar contributions within the RG corresponding to the perimeter and corner renormalisations known from the physics literature, and to prove ultraviolet convergence after their extraction.

1 Current Results

We work at fixed finite physical volume with a compact connected simple gauge group. We allow an arbitrary but positive fine lattice spacing $\varepsilon > 0$ and a sufficiently small coupling g_ε. We perform the first RG step starting from the bare action to a lattice with spacing $L\varepsilon$, for L sufficiently large, independent of ε.

Integrating the fine fluctuations of a Wilson loop gives the schematic expansion

$$W_{\text{fine}}(C) \approx \kappa_C W_{\text{coarse}}(C) + \Delta_C \ , \qquad \kappa_C = e^{-\delta m\, P(C)} z_c^{\,4} \ .$$

[1] Lack of convergence does not contradict Bałaban's UV stability result, nor would it disprove the existence of the physically relevant part of a gauge theory in four dimensions.

For a square contour C, $P(C)$ is its perimeter, δm the perimeter counterterm, and z_c the geometric mean of its four corner factors. Extracting κ_C separates the accumulating scalar contributions from the coarse field dependence. Renormalisation means dividing by this factor and controlling the remaining correction through further steps.

We show that δm and z_c may be computed by integrating the fine fluctuations around the trivial coarse field, the identity, and that they satisfy the following bounds. Up to logarithmic factors in the bare coupling, we have

$$|\delta m| \lesssim \frac{g_\varepsilon^2}{\varepsilon}\,, \qquad\qquad |\log z_c| \lesssim g_\varepsilon^2\,.$$

Each coarse segment contributes a small amount to the perimeter exponent, but refining the lattice increases their number: the counterterm can grow at the rate ε^{-1}. The corner logarithm has no such length factor, although corner contributions can accumulate over RG steps as well. This is the expected ultraviolet scaling of the perimeter and corner contributions.

Extending these estimates to successive RG steps, together with bounds on the remainder, will allow us to control their accumulation under the RG. Using Bałaban's large-field decomposition and forest expansions, we show that corrections to the identity-background calculation gain an additional factor of the coarse-field curvature in small-field regions, while large-field contributions are suppressed by their small probabilities. We also obtain analogous first-step bounds for the correlation function of two square loops with no common vertices, with additional cross terms that decay exponentially with their separation.

However, our current bound on the remainder, up to logarithmic corrections, is only

$$|\mathbb{E}[\Delta_C]| \lesssim \frac{g_\varepsilon^3}{\varepsilon}$$

due to a crude per bond estimate. This is insufficient for concluding UV convergence and we are currently working on eliminating the ε^{-1} term and obtaining the correct g_ε^3 bound.

Further, $W_{\text{coarse}}(C)$ is in fact a "dressed" Wilson loop which retains parts of the large field history in the spirit of Bałaban's large field mechanism. The available first-step comparison costs order g_ε per coarse contour bond, up to logarithmic factors. Since the number of these bonds grows like ε^{-1}, we will retain the dressed insertion through the RG and postpone comparison with an ordinary loop until the final IR scale.

2 Iterating the RG and the Ultraviolet Limit

To go beyond the first-step estimates, we must extend them from the bare lattice measure to the effective measures generated along the RG flow. This requires bounds on the effective interactions and Wilson loop corrections produced at earlier steps, so that the perimeter and corner factors can be extracted at every scale, while the remaining corrections continue to satisfy uniform bounds.

Once we have proved that our Wilson loop estimates can be iterated through arbitrarily many RG steps, we aim to compare theories with successive ultraviolet cutoffs, with their bare couplings tuned to the same renormalised coupling at a fixed infrared scale. The required result is a summable bound on successive differences of the renormalised loop expectations.

3 Research Drafts still under Construction

Our least refined drafts give bounds uniform in the ultraviolet cutoff and convergence along subsequences at a fixed infrared scale, subject to the stated inputs for the large-field steps and the relative-weight comparison. More precisely, the remaining milestones (linked to our corresponding drafts) are:

- Verify our RG induction for the renormalised Wilson loop insertion, with bounds uniform in the number of RG steps, including the large-field renormalisation and locality estimates.
- Verify our bound on the relative weights of the large-field and small-field contributions to the partition function at the fixed infrared scale. From this we have deduced uniform bounds and subsequential convergence of the renormalised Wilson loop expectations.
- Complete and verify the comparison of RG flows at successive ultraviolet cutoffs, with the same renormalised coupling at a fixed reference scale. The latest draft develops a proposed one-step stability estimate, including coupling and large-field feedbacks, together with candidate lattice-refinement bounds for the background fields and fluctuation covariances.

- Use these estimates to prove a summable bound on the differences between renormalised Wilson loop expectations at successive ultraviolet cutoffs, including all remainder terms, and hence convergence along the full sequence.

Assuming structural soundness of our research programme, we expect these milestones should be completed within the next few days. Finally, we conjecture if a method for resolving the mass gap problem exists within or close to the existing literature, it likely lies in finding a correlation function inequality for the correlation functions of renormalised Wilson loops.

4 Concurrent Efforts

A related attempt via construction of correlation functions of gauge invariant objects is being led by Ethan Sussman[2] and Levent Alpoge. To the best of our limited assessment at the time of this announcement, their effort focuses on renormalising pointwise observables of the type $\operatorname{tr} F^2$, with a public note showing the existence of the correlation functions of this composite operator on non-coinciding points for three-dimensional Yang Mills.

We cannot yet comment on the approach taken in the recently announced OpenAI result, since OpenAI has not yet made their output publicly available. However, due to the structural limitations of gauge theories requiring one to apply gauge transformations at each step of the RG flow, it is possibly similar to our approach. The cost of their attempt is similarly inestimable currently.

[2]https://github.com/ethanwsussman/Yang–Mills-LLM-experiment